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Student Number

2008

**TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1-8
- All questions are of equal value

Subject Teachers

Mr I Bradford
Mr M Vuletich

This paper MUST NOT be removed from the examination room

Number of Students in Course: 40

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Total marks – 120

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Marks

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\int_0^1 x e^{-x^2} dx$. 2

(b) Using the substitution $u = e^x$, or otherwise, find $\int \frac{e^x dx}{\sqrt{1-e^{2x}}}$. 2

(c) Find $\int \frac{4x^3 - 2x^2 + 1}{2x - 1} dx$. 3

(d) (i) Find constants a , b and c such that

$$\frac{x^2 + 2x}{(x^2 + 4)(x - 2)} = \frac{ax + b}{x^2 + 4} + \frac{c}{x - 2}.$$
2

(ii) Hence, find $\int \frac{x^2 + 2x}{(x^2 + 4)(x - 2)} dx$. 2

(e) Show, using integration by parts, that 4

$$\int_0^{\frac{\pi}{3}} x \sec^2 x \, dx = \frac{\pi\sqrt{3}}{3} - \ln 2.$$

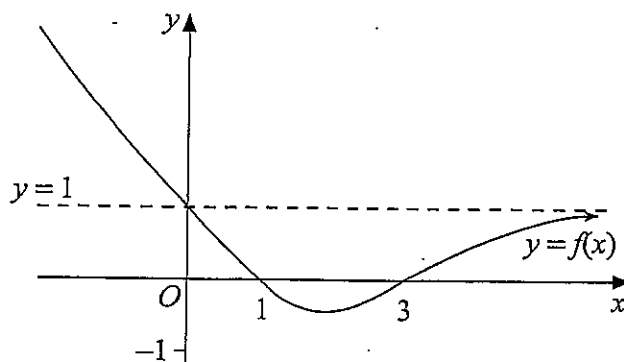
Question 2 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Express $\sqrt{3} - i$ in modulus-argument form. 4
- (ii) Hence evaluate $(\sqrt{3} - i)^6$.
- (b) (i) Simplify $(2i)^3$. 2
- (ii) Hence find all complex numbers z such that $z^3 = 8i$. 2
Express your answers in the form $x + iy$.
- (c) Sketch the region where the inequalities $|z - 3 + i| \leq 5$ and $|z + 1| \leq |z - 1|$ both hold. 3
- (d) Let $w = \frac{3 + 4i}{5}$ and $z = \frac{5 + 12i}{13}$, so that $|w| = |z| = 1$.
- (i) Find wz and $\overline{wz}$ in the form $x + iy$. 2
- (ii) Hence find two distinct ways of writing 65^2 as the sum of $a^2 + b^2$, where 2
 a and b are integers and $0 < a < b$.

Question 3 (15 marks) Use a SEPARATE writing booklet.

- (a) Sketch, without using calculus, the curve $y = \frac{4x^2}{x^2 - 9}$ showing all asymptotes. 3

(b)



The diagram shows the graph of the $y = f(x)$. The graph has a horizontal asymptote at $y = 1$.

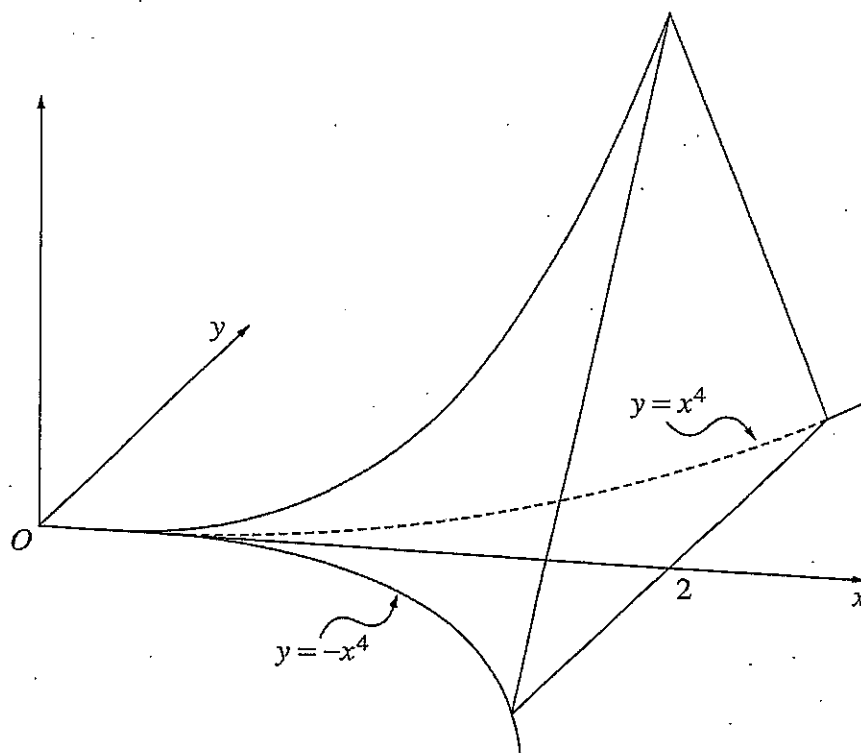
Draw separate one-third page sketches of the graphs of the following:

- (i) $y = |f(x)|$ 2
- (ii) $y = \frac{1}{f(x)}$ 2
- (iii) $y = \ln f(x)$. 2
- (c) Find the equation, in general form, of the tangent to the curve defined by $x^2 - xy + y^3 = 5$ at the point $(2, -1)$. 2

Question 3 continues on page 6

Question 3 (continued)

- (c) The base of a solid is the region in the xy plane enclosed by the curves $y = x^4$, $y = -x^4$ and the line $x = 2$. Each cross-section perpendicular to the x -axis is an equilateral triangle.



- (i) Show that the area of the triangular cross-section at $x = h$ is $\sqrt{3}h^8$. 2
- (ii) Hence find the volume of the solid. 2

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) The ellipse E has Cartesian equation $\frac{x^2}{4} + \frac{y^2}{3} = 1$.
- (i) Write down its eccentricity, the coordinates of its foci S and S' and the equation of each directrix, where S lies on the positive side of the x -axis. 3
- (ii) Sketch E clearly labeling all essential features. 2
- (iii) If P lies on E , then prove that the sum of the distances PS and PS' is independent of P . 2
- (b) $P\left(p, \frac{1}{p}\right)$ and $Q\left(q, \frac{1}{q}\right)$ are two variable points on the rectangular hyperbola $xy = 1$. 2
- If M is the midpoint of the chord PQ and OM is perpendicular to PQ , express q in terms of p .
- (c) (i) Suppose the polynomial $P(x)$ has a double root $x = \alpha$. 2
- Prove that $P'(x)$ also has a root at $x = \alpha$.
- (ii) The polynomial $A(x) = x^4 + ax^2 + bx + 36$ has a double root at $x = 2$. 2
- Find the values of a and b .
- (iii) Factorise the polynomial $A(x)$ of part (ii) over the real numbers. 2

Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) A solid is formed by rotating the region bounded by the curve $y = x(x-1)^2$, the line $y = 0$ and between $x = 0$ and $x = 1$. 3

Use the method of cylindrical shells to find the exact volume of this solid.

- (b) The region between the curve $y = \sin x$ and the line $y = 1$, from $x = 0$ to $x = \frac{\pi}{2}$, is rotated around the line $y = 1$. 4

Using a slicing technique find the exact volume formed.

- (c) A particle is moving in the positive direction along a straight line in a medium that exerts a resistance to motion proportional to the cube of the velocity. No other forces act on the particle, that is, $\ddot{x} = -kv^3$, where k is a positive constant.

At time $t = 0$, the particle is at the origin and has velocity U . At time $t = T$, the particle is at $x = D$ and has velocity V .

- (i) Using the identity $\ddot{x} = \frac{dv}{dt}$ show that 3

$$\frac{1}{V^2} - \frac{1}{U^2} = 2kT.$$

- (ii) Using the identity $\ddot{x} = v \frac{dv}{dx}$, show that 3

$$\frac{1}{V} - \frac{1}{U} = kD.$$

- (iii) Hence show that $\frac{D}{T} = \frac{2UV}{U+V}$. 2

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Graph $y = \ln x$ and draw the tangent to the graph at $x = 1$. 1
- (ii) By considering the appropriate area under the tangent, deduce that 2

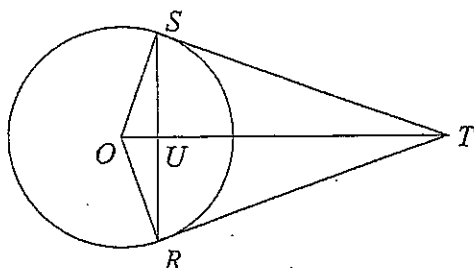
$$\int_1^{\frac{3}{2}} \ln x \, dx \leq \frac{1}{8}.$$

- (b) A mass of 2 kg, on the end of a string 0.6 metres long, is rotating as a conical pendulum, with angular velocity 3π radians per second. The acceleration due to gravity is 10 m/s^2 .

Let θ be the angle that the string makes with the vertical.

- (i) Draw a diagram showing all forces acting on the mass. 1
- (ii) By resolving all forces show that the tension in the string is $10.8\pi^2$ 3
- (iii) Hence, or otherwise, find θ correct to the nearest degree. 1
- (c) Solve for x : $\tan^{-1} 5x - \tan^{-1} 3x = \tan^{-1} \frac{1}{4}$. 3

(d)



The points R and S lie on a circle with centre O and radius 1. The tangents to the circle at R and S meet at T . The lines OT and RS meet at U , and are perpendicular. 4

By considering $\triangle SOU$ and $\triangle TOS$, show that

$$OU \times OT = 1.$$

Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) Let x be a fixed, non-zero number satisfying $x > -1$. 3

Use the method of mathematical induction to prove that

$$(1+x)^n > 1+nx \text{ for } n=2, 3, \dots$$

- (ii) Deduce that $\left(1 - \frac{1}{2n}\right)^n > \frac{1}{2}$ for $n=2, 3, \dots$ 1

- (b) (i) Differentiate $\sin^{-1}(u) - \sqrt{1-u^2}$. 2

- (ii) Hence show that 1

$$\int_0^\alpha \left(\frac{1+u}{1-u}\right)^{\frac{1}{2}} du = \sin^{-1} \alpha + 1 - \sqrt{1-\alpha^2} \text{ for } 0 < \alpha < 1.$$

- (c) A ball of mass 2 kilograms is thrown vertically upward from the origin with an initial speed of 8 metres per second. The ball is subject to a downward gravitational force of 20 newtons and air resistance of $(v^2/5)$ newtons in the opposite direction to the velocity, v metres per second.

Hence, until the ball reaches its highest point, the equation of motion is:

$$\ddot{y} = -\frac{v^2}{10} - 10 \text{ where } y \text{ metres is its height.}$$

- (i) Using the fact that $\ddot{y} = v \frac{dv}{dy}$, show that, while the ball is rising, 3

$$v^2 = 164e^{-\frac{y}{5}} - 100$$

- (ii) Hence find the exact maximum height reached. 1

- (iii) Using the fact that $\ddot{y} = \frac{dv}{dt}$, find how long the ball takes to reach this maximum height. 2

- (iv) How fast is the ball travelling when it returns to the origin? 2

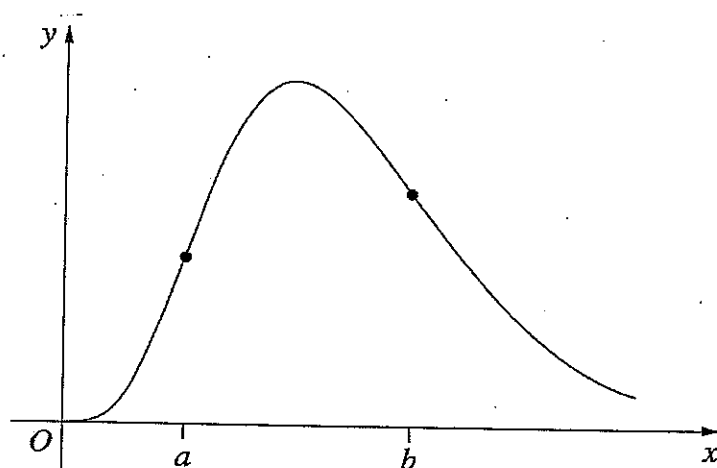
Question 8 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Show that $(1-x^2)^{\frac{n-3}{2}} - (1-x^2)^{\frac{n-1}{2}} = x^2 (1-x^2)^{\frac{n-3}{2}}$. 1

(ii) Let $I_n = \int_0^1 (1-x^2)^{\frac{n-1}{2}} dx$ where $n=0, 1, 2, \dots$ 3

Show that $nI_n = (n-1)I_{n-2}$ for $n=2, 3, 4, \dots$

(b) For $x > 0$, let $f(x) = x^n e^{-x}$, where n is an integer and $n \geq 2$. 4



The two points of inflexion of $f(x)$ occur at $x=a$ and $x=b$, where $0 < a < b$.

Find a and b in terms of n .

(c) A straight line is drawn through a fixed point $P(a, b)$ in the first quadrant on a number plane. The line cuts the positive part of the x -axis at A and the positive part of the y -axis at B .

(i) If $\angle OAB = \theta$, prove that the length of AB is given by $AB = a \sec \theta + b \operatorname{cosec} \theta$. 2

(ii) Show that the length of AB will be a minimum if $\cot \theta = \left(\frac{a}{b}\right)^{\frac{1}{3}}$. 3

(iii) Show that the minimum length of AB is $\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{3}{2}}$. 2

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

KNOX Trial Ext. 2

2008 → Year 12 Extension 2 Trial HSC

Q1

$$\begin{aligned} a) \int_0^1 x e^{-x^2} dx &= -\frac{1}{2} \int_0^1 -2x e^{-x^2} dx \\ &= -\frac{1}{2} \left[e^{-x^2} \right]_0^1 \checkmark \\ &= -\frac{1}{2} (e^{-1} - e^0) \\ &= \frac{1}{2} \left(1 - \frac{1}{e} \right) \\ &= \frac{e-1}{2e} \checkmark \end{aligned}$$

$$b) u = e^x, \quad du = e^x dx$$

$$\begin{aligned} \int \frac{e^x}{\sqrt{1-e^{2x}}} dx &= \int \frac{1}{\sqrt{1-u^2}} du \checkmark \\ &= \sin^{-1} u + C \\ &= \sin^{-1}(e^x) + C \checkmark \end{aligned}$$

$$\begin{aligned} c) \int \frac{4x^3 - 2x^2 + 1}{2x-1} dx &= \int \frac{2x^2(2x-1) + 1}{2x-1} dx \checkmark \\ &= \int 2x^2 + \frac{1}{2x-1} dx \checkmark \\ &= \frac{2x^3}{3} + \frac{1}{2} \ln|2x-1| + C \checkmark \end{aligned}$$

$$\begin{aligned} d) (i) x^2 + 2x &= (ax+b)(x-2) + c(x^2+4) \quad (1) \\ &= (a+c)x^2 + (b-2a)x + 4c-2b \\ &\quad a+c=1 \quad (2) \end{aligned}$$

$$\begin{aligned} \text{Sub. } x=2 \text{ in } (1) &\rightarrow 8=8c \quad b-2a=2 \quad (3) \\ \therefore c=1 \quad \checkmark \end{aligned}$$

$$\begin{aligned} \text{Sub. } c=1 \text{ in } (2) &\rightarrow a=0 \\ \text{Sub. } a=0 \text{ in } (3) &\rightarrow b=2 \end{aligned} \quad \checkmark$$

$$\begin{aligned} (ii) \int \frac{x^2+2x}{(x^2+4)(x-2)} dx &= \int \frac{2}{x^2+4} + \frac{1}{x-2} dx \checkmark \\ &= 2 \left(\frac{1}{2} \tan^{-1} \frac{x}{2} \right) + \ln|x-2| + C \\ &= \tan^{-1} \frac{x}{2} + \ln|x-2| + C \checkmark \end{aligned}$$

$$\begin{aligned} e) u &= x & dv &= \sec^2 x dx \\ du &= dx & v &= \tan x dx \end{aligned}$$

$$\begin{aligned} \int_0^{\frac{\pi}{3}} x \sec^2 x dx &= \left[x \tan x \right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} \tan x dx \checkmark \\ &= \frac{\pi}{3} \tan \frac{\pi}{3} - \int_0^{\frac{\pi}{3}} \frac{\sin x}{\cos x} dx \checkmark \\ &= \frac{\pi\sqrt{3}}{3} + \left[\ln(\cos x) \right]_0^{\frac{\pi}{3}} \checkmark \\ &= \frac{\pi\sqrt{3}}{3} + \left(\ln \frac{1}{2} - \ln 1 \right) \\ &= \frac{\pi\sqrt{3}}{3} - \ln 2 \quad \checkmark \end{aligned}$$

Q2

9) (i) $r = \sqrt{(\sqrt{3})^2 + (-1)^2}$
 $= \sqrt{4}$
 $= 2$ ✓

$\tan \theta = -\frac{1}{\sqrt{3}}$
 $\theta = -\frac{\pi}{6}$

$\therefore \sqrt{3} - i = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)$
 $= 2\left(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)$ ✓

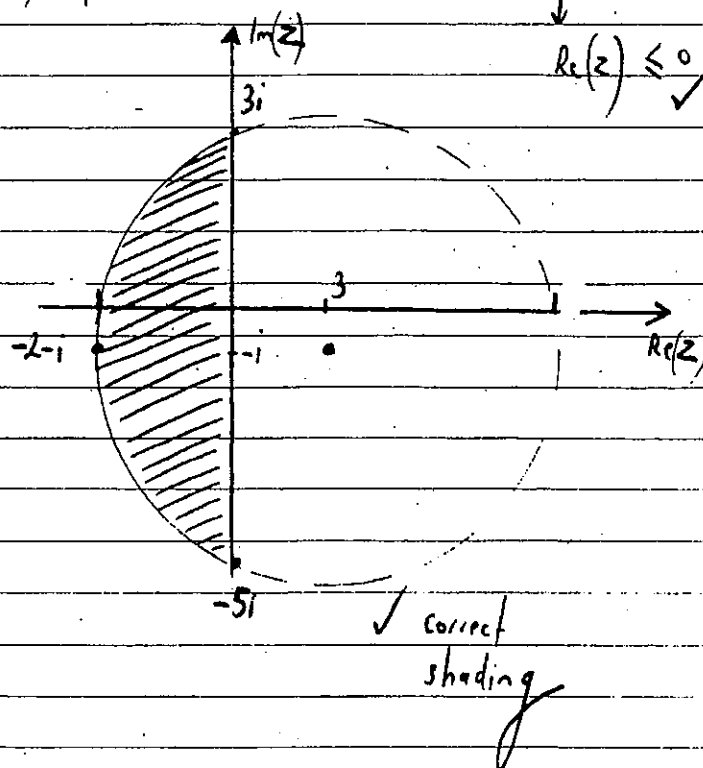
(ii) $(\sqrt{3} - i)^6 = 2^6 \operatorname{cis}\left(-\frac{\pi}{6} \times 6\right)$ ✓
 $= 2^6 (\cos \pi - i \sin \pi)$
 $= -2^6$
 $= -64$ ✓

b) (i) $(-2i)^3 = -8i^3$
 $= 8i$ ✓

(ii) $z^3 = 8i$
 $z^3 - 8i = 0$
 $z^3 + 8i^3 = 0 \rightarrow z^3 + (2i)^3 = 0$
 $(z + 2i)(z^2 - 2iz + 4i^2) = 0$
 $(z + 2i)(z^2 - 2iz - 4) = 0$

$z = -2i$ or $z = \frac{2i \pm \sqrt{4i^2 + 16}}{2}$
 $= \frac{2i \pm \sqrt{-4 + 16}}{2}$
 $= \frac{2i \pm \sqrt{12}}{2}$
 $= i \pm \sqrt{3}$ ✓

c) $|z - 3 + i| \leq 5$ and $|z + i| \leq |z - i|$



d) (i) $wz = \frac{(3 + 4i)(5 + 12i)}{65}$
 $= \frac{-33 + 56i}{65}$
 $= \frac{-33}{65} + \frac{56i}{65}$ ✓

$w\bar{z} = \frac{(3 + 4i)(5 - 12i)}{65}$
 $= \frac{63 - 16i}{65}$
 $= \frac{63}{65} - \frac{16i}{65}$ ✓

(ii) $|wz| = \sqrt{\left(\frac{33}{65}\right)^2 + \left(\frac{56}{65}\right)^2}$
 $= |w| \cdot |z|$

But $1 = \sqrt{\left(\frac{33}{65}\right)^2 + \left(\frac{56}{65}\right)^2}$
 $1 = \left(\frac{33}{65}\right)^2 + \left(\frac{56}{65}\right)^2$
 $\therefore 65^2 = 33^2 + 56^2 \quad \begin{pmatrix} a=33 \\ b=56 \end{pmatrix}$ ✓

Q2 cont'd

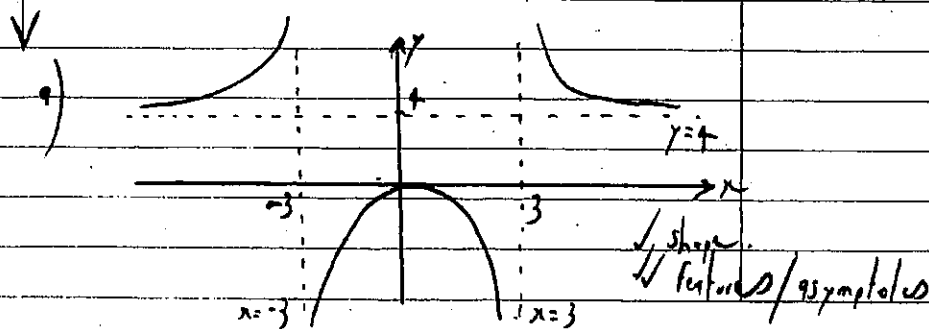
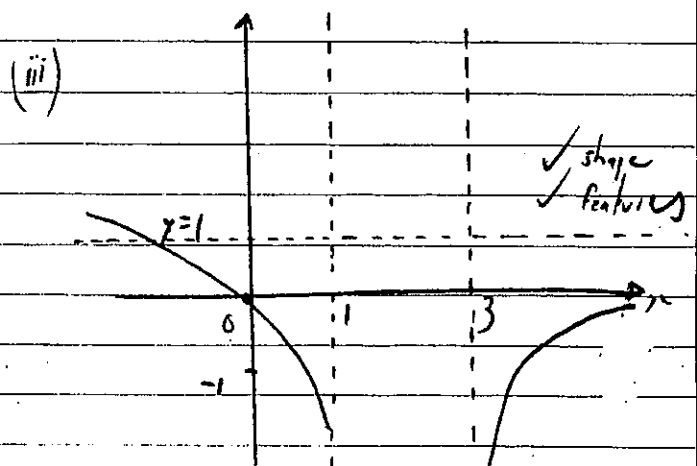
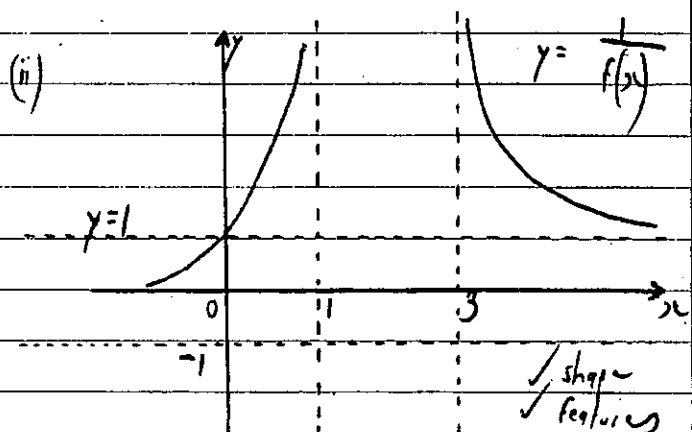
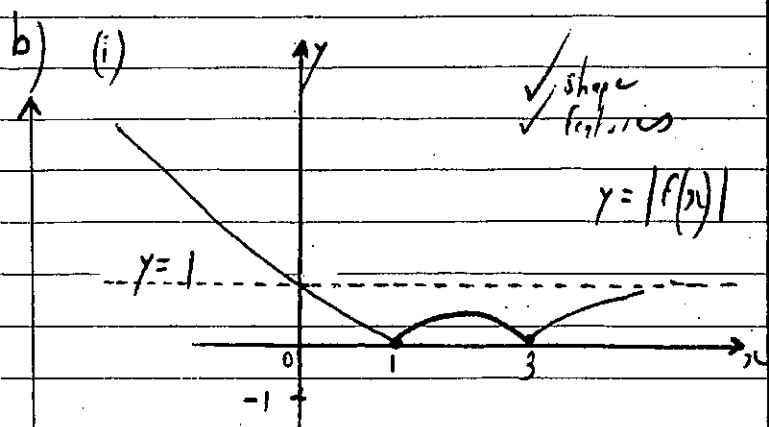
$$\text{Also } |w\bar{z}| = \sqrt{\left(\frac{63}{65}\right)^2 + \left(\frac{16}{65}\right)^2}$$
$$= |w| \cdot |\bar{z}|$$

$$\text{But } |w| = |z| = 1 \text{ and } |\bar{z}| = |z|$$

$$\therefore 1 = \sqrt{\left(\frac{63}{65}\right)^2 + \left(\frac{16}{65}\right)^2}$$

$$\therefore 65^2 = 16^2 + 63^2 \quad \left(\begin{array}{l} a=16 \\ b=63 \end{array} \right) \checkmark$$

Q3



c)

$$x^2 - xy + y^3 = 5$$

$$2x - y - x \frac{dy}{dx} + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{y - 2x}{3y^2 - x} \quad \checkmark$$

At $(2, -1)$, $\frac{dy}{dx} = \frac{-1 - 4}{3 - 2}$

$$= -5$$

= tangent

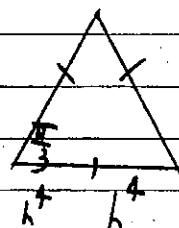
$\therefore y + 1 = -5(x - 2)$

$$y = -5x + 9$$

or

$$5x + y - 9 = 0 \quad \checkmark$$

d) (i)



$$\text{Area} = \frac{1}{2} (2h^4)^2 \sin \frac{\pi}{3} \quad \checkmark$$

$$= \frac{1}{2} (4h^8) \cdot \frac{\sqrt{3}}{2} \quad \checkmark$$

$$= \sqrt{3} h^8$$

(ii)

$$\text{Volume} = \int_0^2 \sqrt{3} x^8 dx$$

$$= \frac{\sqrt{3}}{9} [x^9]_0^2 \quad \checkmark$$

$$= \frac{512\sqrt{3}}{9} \text{ units}^3 \quad \checkmark$$

Q4

9) (i) $\frac{x^2}{2^2} + \frac{y^2}{(\sqrt{3})^2} = 1$

$$b^2 = a^2(1 - e^2)$$

$$3 = 4(1 - e^2)$$

$$e^2 = \frac{1}{4}$$

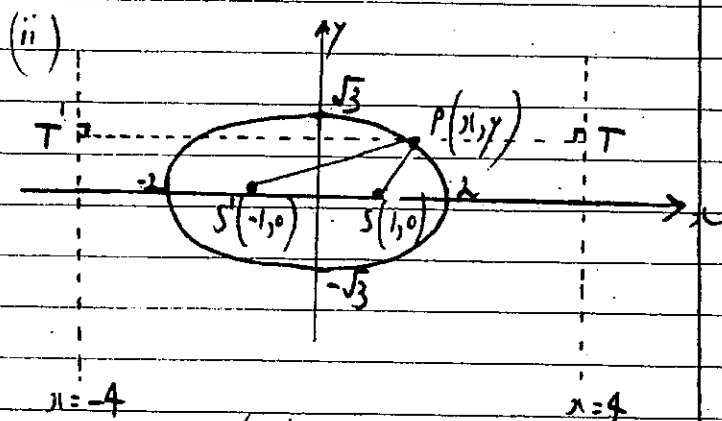
$$e = \frac{1}{2} \checkmark$$

$$S = (\pm ae, 0)$$

ie $S = (1, 0)$ and $S' = (-1, 0) \checkmark$

Directrix at $x = \pm \frac{a}{e}$

ie $x = \pm 4 \checkmark$

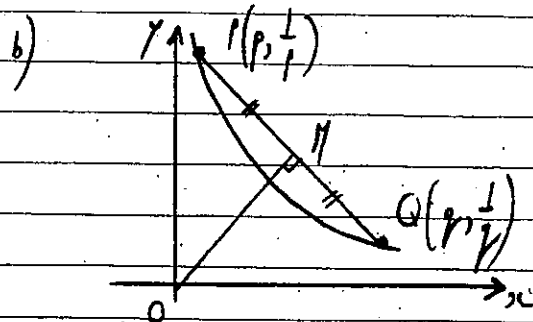


✓ Shape
✓ Features

(iii) Now $PS = \frac{1}{2}PT$
 $= \frac{1}{2}(4 - x)$

and $PS' = \frac{1}{2}PT'$
 $= \frac{1}{2}(4 + x)$

$\therefore PS + PS' = \frac{1}{2}(4 - x) + \frac{1}{2}(4 + x) \checkmark$
 $= 4$ which is independent of P



$$M_{PQ} = \left(\frac{\frac{1}{2} + \frac{1}{2}}{2}, \frac{\frac{1}{2} + \frac{1}{2}}{2} \right)$$

$$= \left(\frac{1}{2}, \frac{1}{2} \right) \checkmark$$

$$m_{PQ} = \frac{\frac{1}{2} - \frac{1}{2}}{\frac{1}{2} - \frac{1}{2}}$$

$$= \frac{0}{0}$$

$$= \frac{1 - 1}{1 - 1}$$

$$= \frac{0}{0}$$

$$= \frac{1}{1}$$

$$= 1 \checkmark$$

since $OM \perp PQ$ then

$$\frac{1}{1} \times -1 = -1 \checkmark$$

$$1 = (1)^2$$

$$\therefore 1 = 1 = 1 \checkmark$$

$$\therefore 1 = 1$$

Q4 con/2

c) (i) If $f(x)$ has a double root at $x=2$

$$\text{Then } f(x) = (x-2)^2 Q(x), Q(x) \neq 0$$

$$\therefore f'(x) = Q(x) \cdot 2(x-2) + (x-2)^2 Q'(x)$$

$$= (x-2) [2Q(x) + (x-2)Q'(x)] \checkmark$$

$$\therefore f'(2) = (2-2) [2Q(2) + (2-2)Q'(2)] \checkmark$$
$$= 0$$

$\therefore x=2$ is also a root of $f'(x)$

(ii) $f(x) = x^4 + ax^2 + bx + 36$

$$f'(x) = 4x^3 + 2ax + b$$

$$f(2) = f'(2) = 0 \text{ since } x=2 \text{ is a double root}$$

$$\therefore 16 + 4a + 2b + 36 = 0$$

$$4a + 2b = -52$$

$$2a + b = -26 \quad (1)$$

$$\text{and } 32 + 4a + b = 0$$

$$4a + b = -32 \quad (2)$$

$$(2) - (1) \rightarrow 2a = -6$$

$$a = -3 \checkmark$$

$$b = -20 \checkmark$$

(iii) $f(x) = x^4 - 3x^2 - 20x + 36$

$$= (x-2)^2 Q(x)$$

$$= (x-2)^2 (kx^2 + Lx + m)$$

By inspection, $m=9, k=1$

$$\text{when } x=1, 1-3-20+36 = (1)(1+L+9)$$

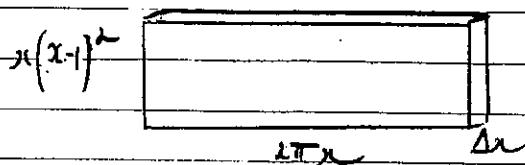
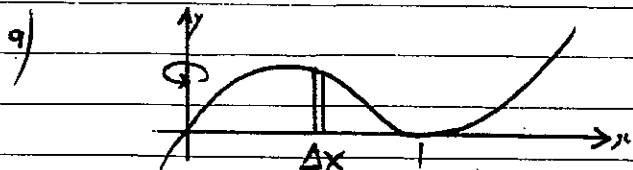
$$14 = 10+L$$

$$4=L$$

$$\therefore f(x) = (x-2)^2 (x^2 + 4x + 9) \checkmark \text{ over } \mathbb{R}$$

or by long division

Q5



$$\Delta V = 2\pi x \cdot x(x-1)^2 \cdot \Delta x$$

$$= 2\pi x^2 (x-1)^2 \cdot \Delta x$$

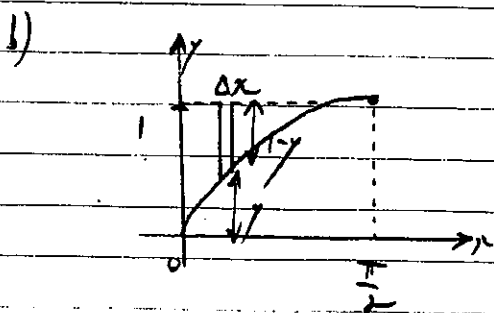
$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^1 2\pi x^2 (x-1)^2 \Delta x \quad \checkmark$$

$$= 2\pi \int_0^1 (x^4 - 2x^3 + x^2) dx$$

$$= 2\pi \left[\frac{x^5}{5} - \frac{2x^4}{4} + \frac{x^3}{3} \right]_0^1 \quad \checkmark$$

$$= 2\pi \left(\frac{1}{5} - \frac{1}{2} + \frac{1}{3} \right)$$

$$= \frac{\pi}{15} \text{ units}^3 \quad \checkmark$$



$$\text{Area each slice} = \pi (1-y)^2$$

$$= \pi (1-\sin x)^2 \quad \checkmark$$

$$\text{Volume each slice} = \pi (1-\sin x)^2 \Delta x$$

$$(\cos 2x = 1 - 2\sin^2 x)$$

$$\therefore V = \pi \int_0^{\frac{\pi}{2}} (1 - 2\sin x + \sin^2 x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} (1 - 2\sin x + \frac{1}{2} - \frac{1}{2} \cos 2x) dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \left(\frac{3}{2} - 2\sin x - \frac{1}{2} \cos 2x \right) dx$$

$$= \pi \left[\frac{3x}{2} + 2\cos x - \frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{2}}$$

$$= \pi \left(\frac{3\pi}{4} - 2 \right) \quad \checkmark$$

$$= \pi \left(\frac{3\pi - 8}{4} \right) \text{ units}^3$$

c) (i) $\frac{dv}{dt} = -kv^3$

$$\frac{dt}{dv} = -\frac{1}{k} v^{-3}$$

$$t = \frac{1}{k} \int v^3 dv$$

$$= -\frac{1}{k} \cdot \frac{1}{2v^2} + C \quad \checkmark$$

$$= \frac{1}{2kv^2} + C$$

At $t=0$, $v=U \rightarrow 0 = \frac{1}{2kU^2} + C$

$$\therefore t = \frac{1}{2k} \left(\frac{1}{v^2} - \frac{1}{U^2} \right) \quad \checkmark$$

At $t=T$, $v=V \rightarrow T = \frac{1}{2k} \left(\frac{1}{V^2} - \frac{1}{U^2} \right)$

$$\frac{1}{V^2} - \frac{1}{U^2} = 2kT \quad \checkmark$$

(ii) $v \frac{dv}{dx} = -kv^3$

$$\frac{dv}{dx} = -kv^2$$

$$\frac{dx}{dv} = -\frac{1}{k} v^{-2}$$

Q5 con'd

$$x = \int -\frac{1}{k} v^2 dv$$
$$= \frac{1}{kv} + C \quad \checkmark$$

At $x=0$, $v=U \rightarrow 0 = \frac{1}{kU} + C \quad \checkmark$

$$\therefore x = \frac{1}{k} \left(\frac{1}{v} - \frac{1}{U} \right)$$

At $x=D$, $v=V \rightarrow D = \frac{1}{k} \left(\frac{1}{V} - \frac{1}{U} \right) \quad \checkmark$

$$\therefore kD = \frac{1}{V} - \frac{1}{U}$$

iii) $\frac{kD}{kT} = \frac{\frac{1}{V} - \frac{1}{U}}{\frac{1}{2} \left(\frac{1}{V^2} - \frac{1}{U^2} \right)}$

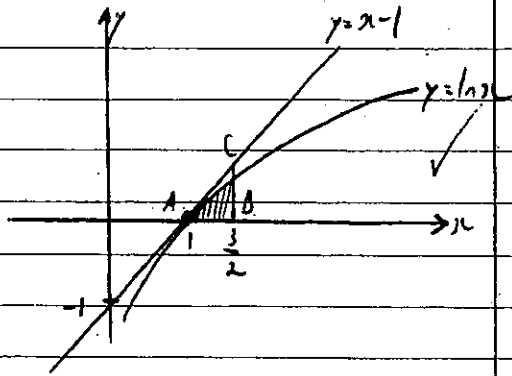
$$= \frac{\frac{U-V}{UV}}{\frac{U^2 - V^2}{2UV^2}}$$

$$= \frac{2UV(U-V)}{(U-V)(U+V)} \quad \checkmark$$

$$\therefore \frac{D}{T} = \frac{2UV}{U+V} \quad \checkmark$$

Q6

(i)



(ii)

$$y = \ln x$$

$$\frac{dy}{dx} = \frac{1}{x}$$

$$\text{At } x=1, \frac{dy}{dx} = 1$$

Tangent is $y = x - 1$ ✓

$$\text{Shaded Area} = \int_1^{3/2} \ln x$$

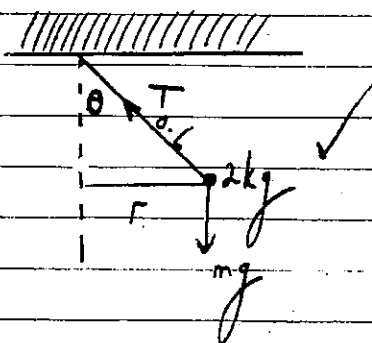
< area of $\triangle ABC$

$$= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \checkmark$$

$$= \frac{1}{8}$$

$$\therefore \int_1^{3/2} \ln x dx < \frac{1}{8}$$

b) (i)



(ii) Vertical $\rightarrow T \cos \theta = 2g$ ✓

Horizontal $\rightarrow T \sin \theta = 2\omega^2 r$

$$\therefore T \cdot \frac{\pi}{0.6} = 2\pi (3\pi)^2 \checkmark$$

$$= 2(0.6) 9\pi^2$$

$$= 10.8\pi^2 \checkmark$$

(iii) $\omega \theta = \frac{20}{T}$

$$= \frac{20}{10.8\pi^2}$$

$$\theta = 79^\circ \checkmark$$

c) $\tan^{-1} 5x - \tan^{-1} 3x = \tan^{-1} \frac{1}{4}$

$$\frac{5x - 3x}{1 + 15x^2} = \frac{1}{4} \checkmark$$

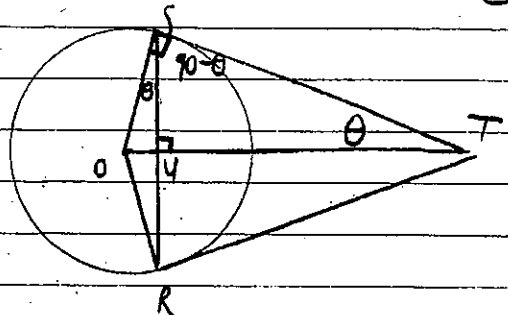
$$1 + 15x^2 = 8x$$

$$15x^2 - 8x + 1 = 0 \checkmark$$

$$(5x - 1)(3x - 1) = 0$$

$$x = \frac{1}{5} \text{ or } \frac{1}{3} \checkmark$$

d)



$$\text{Let } \angle STU = \theta$$

$$\angle OTS = 90^\circ \quad (OT \perp SR)$$

$$\therefore \angle OST = 90^\circ - \theta \quad (\text{L sum of } \triangle SUT)$$

$$\angle OST = 90^\circ \quad (\text{radius } OS \perp \text{tangent } ST)$$

$$\therefore \angle OSU = \theta$$

$$\angle OSU = 90^\circ - \theta \quad (\text{L sum of } \triangle SOUT)$$

$$\therefore \triangle SOUT \parallel \triangle TOS \quad (\text{equiangular}) \checkmark$$

$$\therefore \frac{OU}{OS} = \frac{OS}{OT} \quad (\text{cor. sides in same ratio})$$

$$OU \cdot OT = OS^2 \checkmark \rightarrow OU \cdot OT = 1 \quad (OS = 1)$$

Q7

a) (i) Step (1) $\rightarrow$ For $n=2$,

$$\text{LHS} = (1+x)^2$$

$$= 1+2x+x^2$$

$$\text{RHS} = 1+2x$$

Since $x^2 > 0$, $1+2x+x^2 > 1+2x$.

$\therefore (1+x)^n > 1+nx$ when $n=2$

Step (2) $\rightarrow$ Assume true for $n=k$

$$\text{i.e. } (1+x)^k > 1+kx$$

Step (3) $\rightarrow$ Prove true for $n=k+1$

$$\text{i.e. } (1+x)^{k+1} > 1+(k+1)x$$

$$\text{LHS} = (1+x)^{k+1}$$

$$= (1+x)(1+x)^k$$

$> (1+x)(1+kx)$ from assumption
and since $x > -1 \rightarrow 1+x > 0$ ✓

$$= 1+kx+x+kx^2$$

$$> 1+(k+1)x, \text{ since } kx^2 > 0 \text{ ✓ (i).}$$

Step (4) $\rightarrow$ If the statement is true for $n=2$ and $n=k+1$

it is true for $n=3, 4, \dots$
by mathematical induction

(ii) Let $x = -\frac{1}{2n}$. If $n \geq 2$

then this satisfies conditions in part (i), $x > -1$.

$$\therefore (1+x)^n > 1+nx$$

$$\therefore \left(1 - \frac{1}{2n}\right)^n > 1 - \frac{n}{2n}$$

$$= \frac{1}{2} \text{ ✓}$$

$$\therefore \left(1 - \frac{1}{2n}\right)^n > \frac{1}{2} \text{ for } n=2, 3, \dots$$

(b) (i) Let $y = \sin^{-1}(u) - \sqrt{1-u^2}$, $-1 \leq u \leq 1$

$$\frac{dy}{du} = \frac{1}{\sqrt{1-u^2}} - \frac{1}{2}(1-u^2)^{-\frac{1}{2}} \cdot 2u \text{ ✓}$$

$$-1 \leq u \leq 1$$

$$= \frac{1}{\sqrt{1-u^2}} - \frac{u}{\sqrt{1-u^2}}$$

$$= \frac{1+u}{\sqrt{1-u^2}} \rightarrow \frac{1+u}{\sqrt{1+u} \cdot \sqrt{1-u}}$$

$$= \sqrt{\frac{1+u}{1-u}} \text{ ✓}$$

$$(i) \int_0^a \sqrt{\frac{1+u}{1-u}} du$$

$$= \left[\sin^{-1} u - \sqrt{1-u^2} \right]_0^a$$

$$= \sin^{-1} a - \sqrt{1-a^2} - (0 - \sqrt{1}) \text{ ✓}$$

$$= \sin^{-1} a - \sqrt{1-a^2} + 1, -1 \leq a \leq 1$$

and therefore for $0 < a < 1$

$$(i). \quad v \frac{dy}{dy} = -10 - \frac{v^2}{10}$$

$$\frac{dv}{dy} = \frac{-10}{v} - \frac{v}{10}$$

$$= -\left(\frac{100+v^2}{10v}\right) \text{ ✓}$$

$$\frac{dy}{dv} = \frac{-10v}{v^2+100}$$

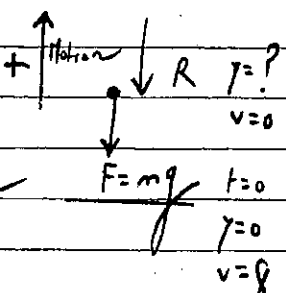
$$y = -10 \int_8^v \frac{v}{v^2+100} dv \text{ ✓}$$

$$= -5 \left[\ln(v^2+100) \right]$$

$$\frac{-y}{5} = \ln\left(\frac{v^2+100}{164}\right)$$

$$\frac{v^2+100}{164} = e^{-\frac{y}{5}} \text{ ✓}$$

$$\therefore v^2 = 164e^{-\frac{y}{5}} - 100$$



Q7 cont'd

c) (ii) Max Height $\rightarrow v=0$

$$y = -5 \ln \left(\frac{100}{1.64} \right) \\ = -5 \ln 1.64 \text{ m} \checkmark$$

(iii) $\frac{dv}{dt} = -10 - \frac{v^2}{10} \rightarrow \frac{100-v^2}{10}$

$$\frac{dt}{dv} = \frac{-10}{v^2+100}$$

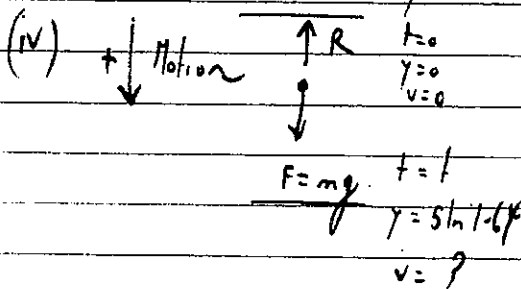
$$t = -10 \int_8^0 \frac{1}{v^2+100} dv$$

$$= 10 \int_0^8 \frac{1}{v^2+100} dv$$

$$= 10 \cdot \frac{1}{10} \left[\tan^{-1} \frac{v}{10} \right]_0^8 \checkmark$$

$$= \tan^{-1} \left(\frac{4}{5} \right) - 0$$

$$\therefore t = \tan^{-1} \left(\frac{4}{5} \right) \text{ seconds} \checkmark$$



$$m\ddot{y} = F - R$$

$$2\ddot{y} = 20 - \frac{v^2}{5}$$

$$\ddot{y} = 10 - \frac{v^2}{10}$$

$$v \frac{dv}{dy} = 10 - \frac{v^2}{10}$$

$$\frac{dv}{dy} = \frac{10}{v} - \frac{v}{10}$$

$$\frac{dv}{dv} = \frac{10v}{100-v^2}$$

$$\left[y \right]_0^{5 \ln 1.64} = 10 \int_0^v \frac{v}{100-v^2} dv$$

$$5 \ln 1.64 = \frac{-5}{10} \left[\ln(100-v^2) \right]_0^v$$

$$- \ln 1.64 = \ln(100-v^2) - \ln 100$$

$$\ln(100-v^2) = \ln \left(\frac{100}{1.64} \right)$$

$$100-v^2 = \frac{100}{1.64}$$

$$v^2 = 100 - \frac{100}{1.64}$$

$$= \frac{64}{1.64}$$

$$v = \frac{8}{\sqrt{1.64}} \text{ since } v > 0$$

$\therefore$ Ball is travelling at 6.25 m/s (2.5 m/s) when returning to origin.

Q8

9) (i) LHS = $\left(1-x^2\right)^{\frac{n-1}{2}} - \left(1-x^2\right)^{\frac{n-1}{2}}$

$$= \left(1-x^2\right)^{\frac{n-1}{2}} \left[1 - \left(1-x^2\right)^{\frac{1}{2}}\right] \checkmark$$

$$= \left(1-x^2\right)^{\frac{n-1}{2}} x^2$$

$$= \text{RHS}$$

Hence shown

(ii) Using IBP,

$$I_n = \int_0^1 \left(1-x^2\right)^{\frac{n-1}{2}} \frac{1}{x} dx$$

$$u = \left(1-x^2\right)^{\frac{n-1}{2}} \quad dv = 1$$

$$du = \frac{n-1}{2} \left(1-x^2\right)^{\frac{n-3}{2}} \cdot -2x dx \quad v = x$$

$$I_n = \int_0^1 \left(1-x^2\right)^{\frac{n-1}{2}} dx$$

$$= \left[x \left(1-x^2\right)^{\frac{n-1}{2}}\right]_0^1 - \frac{n-1}{2} \int_0^1 -2x^2 \left(1-x^2\right)^{\frac{n-3}{2}} dx \checkmark$$

$$= (n-1) \int_0^1 x^2 \left(1-x^2\right)^{\frac{n-3}{2}} dx$$

$$= (n-1) \int_0^1 \left(1-x^2\right)^{\frac{n-3}{2}} dx - (n-1) \int_0^1 \left(1-x^2\right)^{\frac{n-1}{2}} dx \checkmark \text{ from (i)}$$

$$I_n = (n-1) I_{n-2} - (n-1) I_n$$

$$\therefore (n-1) I_n + I_n = (n-1) I_{n-2}$$

$$\therefore n I_n = (n-1) I_{n-2}$$

Hence shown

b) $f(x) = x^n e^{-x}$

$$f'(x) = \left(e^{-x}\right) (nx^{n-1}) + (x^n) (-e^{-x}) \checkmark$$

$$= x^{n-1} e^{-x} (n-x)$$

Stationary point occurs between 2 points of inflexion at $x=n$

$$f''(x) = \left[e^{-x} (n-x)\right]^{n-1} x +$$

$$x^{n-1} \cdot \left[(n-x) e^{-x} + e^{-x} (-1)\right] \checkmark$$

$$= \left[ne^{-x} - xe^{-x}\right]^{n-1} x$$

$$+ x^{n-1} \cdot \left[-ne^{-x} + xe^{-x} - e^{-x}\right]$$

$$= n(n-1) x^{n-2} e^{-x} - (n-1) e^{-x} x$$

$$= nx^{n-1} e^{-x} + x^n e^{-x} - e^{-x} x$$

$$= n(n-1) x^{n-2} e^{-x} - 2nx^{n-1} e^{-x} + x^n e^{-x}$$

$$= e^{-x} x^{n-2} [n(n-1) - 2nx + x^2]$$

$$= e^{-x} x^{n-2} [x^2 - 2nx + n(n-1)]$$

$$= 0 \text{ when } x^2 - 2nx + n(n-1) = 0$$

$$\text{since } a, b > 0 \quad (x-n)^2 = n$$

$$x-n = \pm \sqrt{n}$$

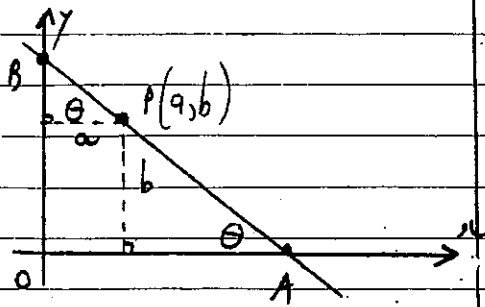
$$x = n \pm \sqrt{n}$$

$$\therefore a = n - \sqrt{n} \text{ and } b = n + \sqrt{n}$$

c) pto

Q8 cont'd

c)



$$\begin{aligned} \text{(i)} \quad \sin \theta &= \frac{b}{AP} \\ AP &= \frac{b}{\sin \theta} \\ &= b \operatorname{cosec} \theta \\ \text{Similarly} \quad \cos \theta &= \frac{a}{PB} \\ PB &= a \sec \theta \\ \therefore AB &= a \sec \theta + b \operatorname{cosec} \theta \end{aligned}$$

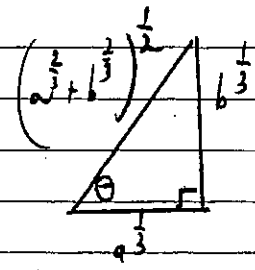
$$\text{(ii)} \quad \frac{d}{d\theta}(AB) = a \sec \theta \tan \theta - b \operatorname{cosec} \theta \cot \theta = 0 \text{ when}$$

$$\begin{aligned} a \sec \theta \tan \theta &= b \operatorname{cosec} \theta \cot \theta \\ \frac{a}{b} &= \frac{\operatorname{cosec} \theta \cot \theta}{\sec \theta \tan \theta} \\ &= \frac{\cot \theta}{\tan^2 \theta} \\ &= \cot^3 \theta \\ \therefore \cot \theta &= \left(\frac{a}{b}\right)^{\frac{1}{3}} \end{aligned}$$

$$\frac{d^2}{d\theta^2} AB = a \sec^3 \theta + a \sec \theta \tan^2 \theta + b \operatorname{cosec} \theta \cot^2 \theta + b \operatorname{cosec}^3 \theta$$

since $0 \leq \theta \leq \frac{\pi}{2}$ and all trig. ratios > 0 then $\frac{d^2}{d\theta^2} AB > 0$ so $\cot \theta = \left(\frac{a}{b}\right)^{\frac{1}{3}}$ minimises AB

$$\text{iii.} \quad \cot \theta = \frac{a^{\frac{1}{3}}}{b^{\frac{1}{3}}} \rightarrow \theta \text{ is acute}$$



$$\begin{aligned} r^2 &= a^{\frac{2}{3}} + b^{\frac{2}{3}} \\ r &= \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}} \\ \therefore \sec \theta &= \frac{\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}}}{a^{\frac{1}{3}}} \\ \operatorname{cosec} \theta &= \frac{\left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}}}{b^{\frac{1}{3}}} \end{aligned}$$

$\therefore$ Minimum length of AB

$$\begin{aligned} &= \frac{a \cdot \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}{a^{\frac{1}{3}}} + \frac{b \cdot \sqrt{a^{\frac{2}{3}} + b^{\frac{2}{3}}}}{b^{\frac{1}{3}}} \\ &= \left(a^{\frac{1}{3}} + b^{\frac{1}{3}}\right)^{\frac{1}{2}} \cdot \left(a^{\frac{2}{3}} + b^{\frac{2}{3}}\right)^{\frac{1}{2}} \\ &= \left(a^{\frac{1}{3}} + b^{\frac{1}{3}}\right)^{\frac{3}{2}} \end{aligned}$$

Hence shown